

# AP Biology: The Complete Exam Guide

All 8 units explained, the six free-response question types, a data-analysis playbook, and 10 hard practice questions with full explanations.

Hi, I'm **Pigo!** I'll drop tips all through this guide.



**3 hrs**

90 min multiple choice  
+ 90 min free response

**60+6**

multiple-choice +  
free-response questions

**8 units**

from molecules  
to ecosystems

**70–71%**

scored 3 or higher  
in recent years

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# 01 THE EXAM AT A GLANCE

## Two sections, one score

Your AP score (1–5) combines a multiple-choice section and a free-response section, each worth half. Both lean heavily on data, graphs and experiments, not just memorized facts.

**50%**

Section I • Multiple choice • 60 questions

**50%**

Section II • Free response • 6 questions

### Exam structure

| Section             | Weight & timing                             | What you do                                                                                                                                                                              |
|---------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I • Multiple choice | 50% of score<br>1 hr 30 min<br>60 questions | Individual questions and sets built around data, diagrams and experiments. You explain concepts, analyze visuals, use the scientific method and do calculations.                         |
| II • Free response  | 50% of score<br>1 hr 30 min<br>6 questions  | 2 long questions (interpreting an experiment; analyzing data with a graph) and 4 short questions (scientific investigation, conceptual analysis, analysis of a model, analysis of data). |

#### HYBRID DIGITAL

##### Click in Bluebook, write on paper

You answer multiple-choice questions and read the free-response questions in the **Bluebook** app, then handwrite your free-response answers in a paper booklet. Practice writing clear, organized work by hand.

#### CALCULATOR + EQUATIONS

##### Bring a calculator

A calculator is allowed on the whole exam, and an **equations and formulas sheet** is provided (chi-square, Hardy-Weinberg, water potential, and more). Practice using it, not memorizing it.

### How to use this guide

#### Just starting?

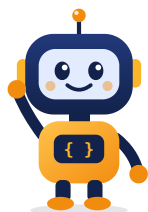
Read pages 2–6, then follow the roadmap on page 8 month by month.

#### Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

#### Exam in weeks?

Take the 10 questions on page 9 in 18 minutes, then use the final-four-weeks plan.



**Pigo's tip:** Most AP Bio points come from **reasoning with evidence**: "the data show X, which means Y because Z". Practice that sentence pattern.

## SECTION I

# 02 The multiple-choice section

60 questions in 90 minutes gives you 1 minute 30 seconds per question.

Data-heavy sets take longer, so move quickly through recall questions.

**60**

questions, many built on data or diagrams

**1:30**

minutes per question on average

**50%**

of your total AP score

**0**

points lost for a wrong answer, so never leave one blank

### How the questions are weighted by unit

College Board gives a range for each unit's share of the exam. Bars show minimum to maximum.



**What this means for your study time:** Natural selection, energetics and gene expression are the biggest units. But every unit shows up, and the free-response section can pull from any of them.

### Strategies that work on this section

- Read the graph before the question: axes, units, the independent variable and the control.
- Use the formula sheet for water potential, chi-square and Hardy-Weinberg.
- Predict before you look at the choices for "what would happen if..." questions.
- Watch for one-word traps: "always", "only", "directly".
- Follow the energy and the electrons in respiration and photosynthesis questions.
- Flag long data sets and return once the quick points are banked.

## 03 WHAT THE COURSE COVERS

# The 8 units, in plain English

Everything on the exam fits into these eight units.

UNIT

1

8–11%

### Chemistry of Life

Water and the molecules of life.

**Be ready to:** explain water's properties; relate structure to function in carbohydrates, lipids, proteins and nucleic acids; explain dehydration synthesis and hydrolysis.

UNIT

2

10–13%

### Cells

Organelles, membranes and transport.

**Be ready to:** link organelles to functions; explain surface-area-to-volume limits; describe membrane transport; calculate water potential; explain endosymbiosis.

UNIT

3

12–16%

### Cellular Energetics

Enzymes, photosynthesis and respiration.

**Be ready to:** explain enzyme activity and inhibition; trace electrons and energy through photosynthesis and cellular respiration; predict effects of blocking a step.

UNIT

4

10–15%

### Cell Communication & Cell Cycle

Signaling and cell division.

**Be ready to:** describe signal transduction pathways; predict effects of mutations in a pathway; explain feedback; describe mitosis, checkpoints and cancer.

UNIT

5

8–11%

### Heredity

How traits pass between generations.

**Be ready to:** explain meiosis and genetic variation; solve Mendelian and non-Mendelian crosses; use pedigrees; calculate recombination frequency; run chi-square tests.

UNIT

6

12–16%

### Gene Expression & Regulation

From DNA to protein, and how genes are switched.

**Be ready to:** describe replication, transcription and translation; explain operons and eukaryotic regulation; predict effects of mutations; explain biotechnology tools.

UNIT

7

13–20%

### Natural Selection

Evolution and its evidence.

**Be ready to:** explain natural selection, genetic drift and gene flow; use Hardy-Weinberg; read phylogenetic trees; describe speciation and evidence of common ancestry.

UNIT

8

10–15%

### Ecology

Organisms, populations and ecosystems.

**Be ready to:** explain behavior and energy flow; model population growth; describe community interactions, biodiversity and the effects of disruptions.

**Lab skills count.** Many questions ask you to identify variables, controls and hypotheses, or to justify a claim with data. Practice these in every unit, not just before May.

# 04 SECTION II

## The 6 free-response questions

90 minutes, 2 long and 4 short questions. The question types are predictable, so you can practice each one on purpose.

|           |           |    |    |    |    |
|-----------|-----------|----|----|----|----|
| Q1 • long | Q2 • long | Q3 | Q4 | Q5 | Q6 |
|-----------|-----------|----|----|----|----|

Long questions are worth more points than short ones

**Q1**

### Interpreting & evaluating experimental results

**Long**

Describe a biological concept, identify parts of an experiment, analyze the data and predict results.

Name the independent variable, dependent variable and control explicitly.

**Q2**

### Experimental results with graphing

**Long**

Same skills as Q1, plus you construct a graph from data, with error bars where asked.

Graph checklist: title, labeled axes with units, correct graph type, scale, key.

**Q3**

### Scientific investigation

**Short**

Describe or design parts of an experiment: hypotheses, variables, controls and justification.

A null hypothesis predicts "no difference" or "no effect".

**Q4**

### Conceptual analysis

**Short**

Explain a biological concept and predict the effect of a change to a system.

Use cause → effect chains: "If X decreases, then Y..., because...".

**Q5**

### Analysis of a model or visual

**Short**

Interpret a diagram or model, explain its limitations, and use it to predict.

Point to features in the model when you justify.

**Q6**

### Analysis of data

**Short**

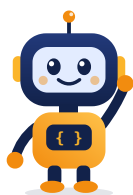
Read a graph or table, describe trends, and support a claim with specific numbers.

Quote values with units: "rose from 2.1 to 6.4 mg/L".

### ◆ A time plan for the 90 minutes

|               |               |           |           |           |           |
|---------------|---------------|-----------|-----------|-----------|-----------|
| Q1<br>~25 min | Q2<br>~25 min | Q3<br>~10 | Q4<br>~10 | Q5<br>~10 | Q6<br>~10 |
|---------------|---------------|-----------|-----------|-----------|-----------|

College Board suggests about 25 minutes for each long question and 10 for each short one. Adjust to your strengths.



Answer each lettered part **separately**. Readers look for each point in its own place.

FREE-RESPONSE PLAYBOOK

05

# Anatomy of a full-credit answer

Here is a short experimental-analysis prompt with a full-credit answer, and the habits that earn points.

PRACTICE TASK • WRITTEN BY ACT SAT GO

**Practice prompt (Q1/Q3 style):** Students measured  $O_2$  production by aquatic plants at four light intensities (0, 50, 100 and 200  $\mu\text{mol photons/m}^2/\text{s}$ ), with five plants per group, at 25°C. (a) Identify the independent and dependent variables. (b) State a null hypothesis. (c) Predict the effect of adding DCMU, a chemical that blocks electron flow out of photosystem II, and justify your prediction.

Full-credit response

- ① (a) Independent variable: light intensity. Dependent variable: rate of  $O_2$  production.
- ② (b) Light intensity has no effect on the rate of  $O_2$  production.
- ③ (c)  $O_2$  production will drop to nearly zero at every light intensity.
- ④ If electrons cannot leave photosystem II, the electron transport chain stops, so P680 cannot pass on electrons and water is no longer split. Water splitting is the source of the  $O_2$  released.

- ① **Name both variables** exactly as measured.
- ② **Null = no effect**, stated about these variables.
- ③ **Make a specific, testable prediction**.
- ④ **Justify with the mechanism**, linking the blocked step to the measured result.

## Five habits for every free-response question

1

Answer every lettered part, in order, with a label.

2

Show the equation, the substitution and the units.

3

Justify with a cause → effect sentence, not a word.

4

Use the reference sheet; you don't need to memorize it.

5

Blank part? Write the relationship you would use. It can earn credit.

# 06 LEARN FROM OTHERS

## 10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

Evolution has a **goal!**  
(It doesn't.)



### 01 Individuals evolving

Populations evolve; individuals don't. Selection acts on existing variation.

### 02 "Needs" in evolution

Organisms don't mutate because they need to. Mutations are random; selection is not.

### 03 Mixing up mitosis and meiosis

Meiosis makes 4 genetically different haploid cells; crossing over and independent assortment create variation.

### 04 Water potential signs

Water moves from higher (less negative) to lower (more negative) water potential. Solute potential is always  $\leq 0$ .

### 05 Enzymes "used up"

Enzymes are not consumed. Denaturation changes shape and the active site, which stops function.

### 06 Graph errors

Missing axis labels, units, a title, or using a line graph for categories all cost points.

### 07 Vague justifications

"It affects the cell" scores nothing. Name the molecule, the step and the result.

### 08 Chi-square misread

If  $\chi^2$  is less than the critical value, you fail to reject the null hypothesis. Use  $df = \text{categories} - 1$ .

### 09 Hardy-Weinberg mix-up

$q^2$  is the recessive phenotype frequency. Take the square root to get  $q$ , then  $p = 1 - q$ .

### 10 Ignoring the control

Compare results to the control group before drawing any conclusion.

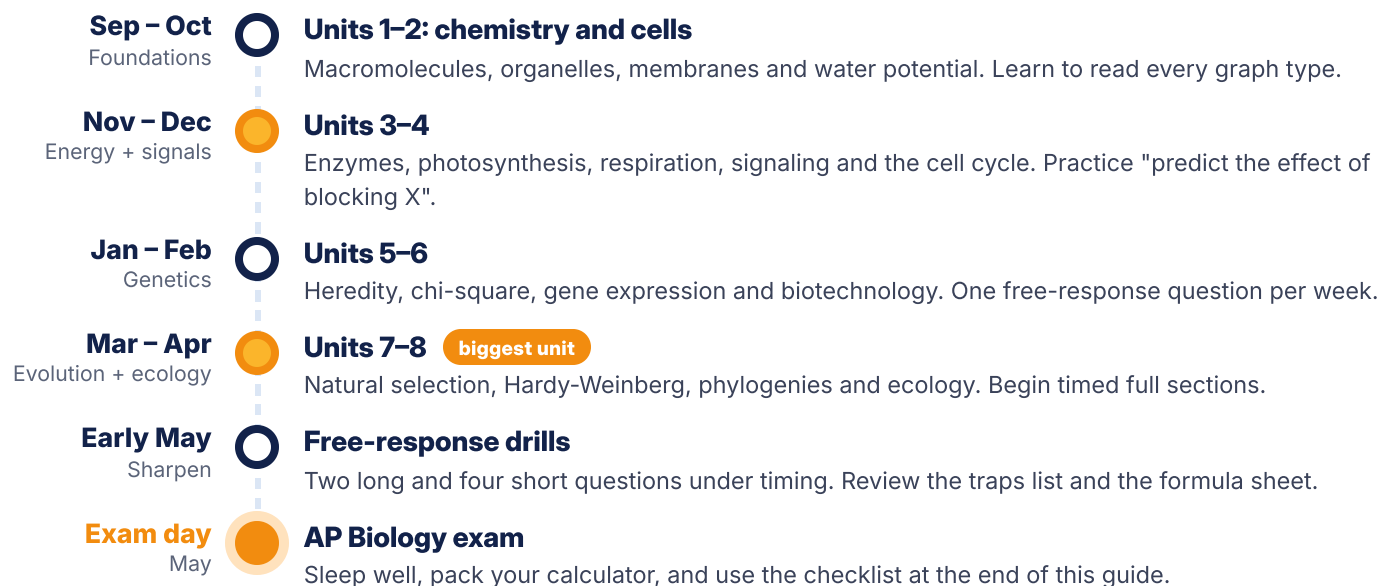
**Quick self-check:** in a population, 16% show a recessive trait. What fraction are heterozygous? ( $q = 0.4$ ,  $p = 0.6$ ,  $2pq = 0.48$ , so 48%.)

07

PLAN YOUR YEAR

# Your study roadmap to May

A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, but keep practicing data analysis throughout.



## ◆ The final four weeks

| Week        | Focus                                                                       | Target                        |
|-------------|-----------------------------------------------------------------------------|-------------------------------|
| 4 weeks out | Diagnose: 30 multiple-choice questions plus one long free-response question | List your three weakest units |
| 3 weeks out | Fix weak units; one of each free-response type                              | 80%+ on unit drills           |
| 2 weeks out | Full 3-hour practice exam                                                   | Finish both sections on time  |
| Final week  | Light review: formula sheet, graphing checklist, traps                      | Rest the night before         |

# 08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Give yourself **18 minutes** for all ten, then check page 11.

## 1 UNIT 2 • WATER POTENTIAL

A 0.2 M sucrose solution is at 22°C (295 K) in an open beaker. Using  $\Psi_s = -iCRT$  with  $R = 0.0831$  L·bar/(mol·K), what is its water potential?

- A +4.9 bar
- B -0.49 bar
- C -9.8 bar
- D -4.9 bar

## 2 UNIT 3 • ENZYMES

An inhibitor reduces an enzyme's reaction rate, but the maximum rate ( $V_{max}$ ) is reached again at very high substrate concentration. What best explains this?

- A The inhibitor competes with the substrate for the active site
- B The inhibitor binds an allosteric site and permanently changes the active site
- C The inhibitor denatures the enzyme
- D The inhibitor is a cofactor the enzyme needs

## 3 UNIT 3 • PHOTOSYNTHESIS

DCMU blocks electron flow from photosystem II to plastoquinone. Which outcome is expected in illuminated chloroplasts?

- A  $O_2$  release increases because more electrons stay in PSII
- B The Calvin cycle speeds up
- C ATP synthesis increases through cyclic flow from PSII
- D  $O_2$  release stops and NADPH production falls

## 4 UNIT 4 • SIGNAL TRANSDUCTION

A mutation makes a G protein unable to hydrolyze GTP. What is the most likely effect on the pathway?

- A The pathway can never be activated
- B The pathway stays switched on even without the signal
- C The receptor can no longer bind its ligand
- D Second messengers are destroyed faster

## 5 UNIT 5 • CHI-SQUARE

A dihybrid cross expected to give a 9:3:3:1 ratio produced these offspring:

| Phenotype | A_B_ | A_bb | aaB_ | aabb |
|-----------|------|------|------|------|
| Observed  | 315  | 108  | 101  | 32   |

What is  $\chi^2$ , and what is the conclusion at  $p = 0.05$  (critical value 7.81)?

- A  $\chi^2 \approx 0.47$ ; fail to reject the null hypothesis
- B  $\chi^2 \approx 0.47$ ; reject the null hypothesis
- C  $\chi^2 \approx 7.9$ ; reject the null hypothesis
- D  $\chi^2 \approx 4.7$ ; fail to reject the null hypothesis

## 6 UNIT 5 • LINKAGE

A testcross of a double heterozygote produced 205 + 205 parental-type offspring and 45 + 45 recombinant-type offspring. How far apart are the two genes?

- A 9 map units
- B 18 map units
- C 22 map units
- D 45 map units

## 08 TEST YOURSELF (continued)

### 7 UNIT 6 • GENE REGULATION

In *E. coli*, a mutation in the lac operator prevents the repressor from binding. What happens to lac gene expression when **lactose is absent**?

- A Expression is off, as normal
- B Expression is on (constitutive)
- C Expression requires glucose to be present
- D The repressor protein is no longer produced

### 8 UNIT 7 • HARDY-WEINBERG

In a population of 500 at Hardy-Weinberg equilibrium, 80 individuals show a recessive phenotype. How many individuals are expected to be heterozygous?

- A 120
- B 160
- C 240
- D 320

### 9 UNIT 8 • ENERGY FLOW

Producers in an ecosystem capture 20,000 kcal of energy. About how much energy is available to **tertiary** consumers?

- A 2,000 kcal
- B 200 kcal
- C 20 kcal
- D 2 kcal

### 10 UNIT 4 • CELL CYCLE

A cell has a nonfunctional p53 protein. Which outcome is most likely after its DNA is damaged?

- A The cell stops at the G1 checkpoint and repairs the DNA
- B The cell undergoes apoptosis immediately
- C The cell stops making cyclins
- D The cell continues dividing with damaged DNA

09

CHECK YOUR WORK

# Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

| #  | Answer       | Why                                                                                                                                                                    |
|----|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | D - -4.9 bar | Sucrose doesn't ionize, so $i = 1$ : $\Psi_s = -(1)(0.2)(0.0831)(295) \approx -4.9$ bar. In an open beaker $\Psi_p = 0$ , so $\Psi = \Psi_s$ . Choice C uses $i = 2$ . |
| 2  | A            | Competitive inhibitors compete for the active site, so enough substrate outcompetes them and $V_{max}$ is unchanged. Noncompetitive inhibition lowers $V_{max}$ .      |
| 3  | D            | With PSII blocked, P680 can't pass electrons on, so water isn't split (no $O_2$ ) and no electrons reach $NADP^+$ . The Calvin cycle slows without NADPH.              |
| 4  | B            | GTP hydrolysis switches a G protein off. If it can't hydrolyze GTP, it stays active and keeps signaling, a common route to uncontrolled division.                      |
| 5  | A            | Expected values (556 total): 312.75, 104.25, 104.25, 34.75. $\chi^2 = \sum(o - e)^2/e \approx 0.47 < 7.81$ (df = 3), so the data fit a 9:3:3:1 ratio.                  |
| 6  | B - 18       | Recombination frequency = $90/500 = 0.18 = 18\%$ . One map unit equals 1% recombination.                                                                               |
| 7  | B            | Without repressor binding, RNA polymerase transcribes the genes whether or not lactose is present: constitutive expression.                                            |
| 8  | C - 240      | $q^2 = 80/500 = 0.16$ , so $q = 0.4$ and $p = 0.6$ . $2pq = 0.48$ , and $0.48 \times 500 = 240$ .                                                                      |
| 9  | C - 20 kcal  | About 10% passes up each level: primary 2,000, secondary 200, tertiary 20 kcal.                                                                                        |
| 10 | D            | p53 normally halts the cycle (or triggers apoptosis) after DNA damage. Without it, damaged cells keep dividing, which can lead to cancer.                              |

## 9–10

Strong. Move on to full timed free-response practice.

## 6–8

On track. Revisit the unit behind every miss on page 4.

## 0–5

Build foundations now. Start with energetics and gene expression (Units 3 and 6).

## 10 KNOW THE TARGET How students score

In recent years, about 70–71% of students earned a 3 or higher on AP Biology, and about 15–19% earned a 5. Steady preparation is what moves a score up a full point.

**70–71%**

of students earned a 3 or higher in recent years

**15–19%**

earned the top score of 5 in recent years

**3+**

is the score many colleges accept for credit or placement

### What each AP score means

**5**

Extremely well qualified

**4**

Very well qualified

**3**

Qualified

**2**

Possibly qualified

**1**

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

### Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, plus **two pencils and a dark-ink pen** for the free-response booklet
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ A **calculator** (four-function, scientific or graphing) with fresh batteries
- ☐ Plan your time: about **90 s** per multiple-choice question; **~25 min** per long and **~10 min** per short free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ Use the **equations and formulas sheet**

**In the free-response section:** answer each lettered part separately, name variables and controls, use specific numbers from the data, and justify every prediction with a biological mechanism.



ACT SAT GO • 1-ON-1 AP TUTORING

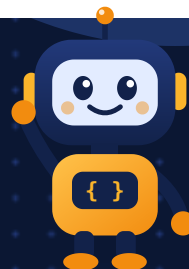
# Aiming for a 5? Get a personal AP Bio plan.

Book a free diagnostic lesson. We'll find your weakest units, review your practice responses, and give you a week-by-week plan to exam day in May.

**Book a free diagnostic lesson →**



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## Concept tutoring

Live 1-on-1 sessions on all eight units, with data-analysis drills after every lesson.

## Free-response coaching

Practice all six question types and get feedback against rubric-style criteria.

## Timed exam practice

Full-length multiple-choice sets with score reports, so you see progress before May.

## ◆ Contact us

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